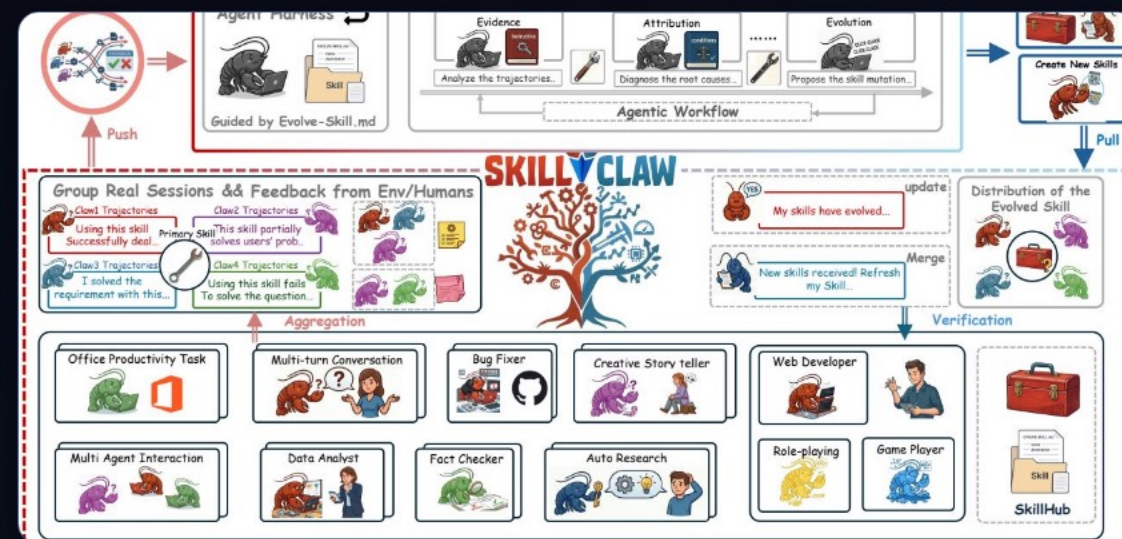


SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

From static, manually installed skills to a closed-loop system where multi-user trajectories drive shared, evidence-grounded skill updates.

Let Skills Evolve Collectively with Agentic Evolver

- Aggregates trajectories from heterogeneous users and tasks.
- Diagnoses root causes through attribution over behavior traces.
- Mutates shared skills conservatively via verified acceptance.



Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

Static Skill Paradigm

- Skills are manually authored and installed once.
- Runtime failures do not systematically update skills.
- Users independently rediscover similar fixes.
- No ecosystem-level mechanism accumulates experience.

Desired Evolving Paradigm

- Interaction trajectories become reusable evidence.
- Skill mutations are proposed from diagnosed causes.
- Validated improvements synchronize to all users.
- System quality compounds through collective use.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

- Stage 1

Evidence

Collect trajectories from multi-user sessions exercising shared skills under diverse contexts.
- Stage 2

Attribution

Analyze success/failure traces to identify likely root causes in skill behavior.
- Stage 3

Evolution

Generate candidate skill mutations via agentic open-ended reasoning.
- Stage 4

Distribution

Validate and synchronize accepted updates back into the shared skill pool.

Multi-User Interaction → Session Collection → Agentic Evolver → Skill Synchronization → (Loop)

The loop converts isolated runtime experiences into collective, continuously improving capability.



Evidence → Attribution → Evolution in Three Stages

- **Evidence Analysis:** ingest trajectory sets with outcomes, action traces, and context.
- **Attribution:** diagnose whether failure stems from missing guidance, brittle logic, or scope gaps.
- **Evolution:** draft candidate skill mutations and expected behavioral effects.
- Workflow is **reasoning-driven**, not fixed rule templates.

Why this matters

- Supports context-aware updates grounded in observed behavior.
- Handles open-ended failure modes across heterogeneous tasks.
- Enables conservative acceptance through downstream validation.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

- Compatible systems include OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw.
- Diverse users invoke the same skill under distinct prompts, tools, and constraints.
- Aggregation reveals complementary success and failure signatures.
- Single-user traces are often insufficient to separate robust improvements from one-off patches.

Group Session Aggregation (Architecture Segment)

User A/B/C sessions feed a shared trajectory buffer. The evolver operates over grouped evidence to estimate where a skill generalizes vs. where it breaks, then proposes mutations with higher transfer potential.

Key effect: richer boundary detection for when and why the current skill fails.

Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Conservative verification retained only 1 of 6 candidate updates.

Date	Outcome	Rationale
2026-04-01	Rejected	Overfit to narrow prompt style; reduced robustness across variants.
2026-04-02	Rejected	Introduced unnecessary branching; higher failure rate in edge tool states.
2026-04-03	Accepted	Improved step ordering and fallback guidance; net reliability gain.
2026-04-04	Rejected	Regressed output consistency for multimodal handoff stage.
2026-04-05	Rejected	Added verbosity without measurable execution benefit.
2026-04-06	Rejected	Conflicted with existing validated constraints in skill pool.

Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Three accepted updates before performance plateaued.

Date	Outcome	Rationale
2026-04-01	Accepted	Added explicit auth fallback path for common remote mismatch.
2026-04-02	Rejected	Candidate duplicated checks already encoded in base skill.
2026-04-03	Accepted	Improved token/SSH decision logic under ambiguous credential states.
2026-04-04	Rejected	Triggered unnecessary retries, increasing latency without success gain.
2026-04-05	Accepted	Refined final fallback sequencing; higher completion on hard cases.
2026-04-06	Rejected	No incremental gain beyond prior accepted version (plateau).

Three Properties That Distinguish SkillClaw from Existing Systems

- **Collective evolution:** individual interactions improve a shared ecosystem.
- **Fully automatic:** no manual curation required for routine updates.
- **Agentic evolution paradigm:** open-ended reasoning drives skill mutation.

Implications

- Continuously improving agent systems become practical at scale.
- Validation-gated updates balance innovation with reliability.
- Multi-user ecosystems convert usage into compounding capability.